

Gradient Descent for Multiple Linear Regression

This section combines the concepts of multiple linear regression and vectorization to create an efficient implementation of the gradient descent algorithm.

Review: Vector Notation 📌

To simplify our equations, we group our parameters and features into **vectors**.

- **Parameters:**

- Instead of separate parameters $w_1, w_2, \dots, w_n$, we define a single parameter **vector** $\vec{w} = [w_1, w_2, \dots, w_n]$.
- b remains a single number (a scalar).

- **Model:** The model can be written compactly using the **dot product**:

$$f_{\vec{w},b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$$

- **Cost Function:** The cost function is now a function of the vector $\vec{w}$ and the scalar b :

$$J(\vec{w}, b)$$

The Gradient Descent Algorithm

The algorithm updates all n of the w parameters and the b parameter simultaneously on each iteration.

- **Update Rule for w_j (for $j=1\dots n$):**

$$w_j := w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b)$$

- **Update Rule for b :**

$$b := b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b)$$

Derivative Formulas for Multiple Regression

The specific derivative terms for the multiple linear regression cost function are very similar to the univariate case.

- **Derivative with respect to w_j :**

$$\frac{\partial}{\partial w_j} J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)}$$

- **Derivative with respect to b :**

$$\frac{\partial}{\partial b} J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)})$$

The main difference from the single-feature model is that we now have a separate update for each feature's weight (w_1, w_2 , etc.), and the derivative for w_j specifically uses the corresponding feature value, x_j .

Alternative Method: The Normal Equation (Side Note)

- For linear regression *only*, there is a non-iterative method for finding the optimal w and b called the **Normal Equation**.
- It involves solving for the parameters directly using advanced linear algebra, without needing to choose a learning rate α or run multiple iterations.
- **Disadvantages:**
 - It **does not generalize** to other algorithms like logistic regression or neural networks.
 - It can be very **slow** if the number of features (n) is large.
- **Conclusion:** Gradient descent is a more general and often more practical method for most machine learning applications.